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Experiment 1

Aim:
To study the basics of computer forensics and explore different tools used in forensic investigations.

Objectives:

1. To understand the fundamentals and importance of digital forensics.
2. To study the various phases of a digital forensic investigation.
3. To understand the international legal framework related to cybercrime.
4. To explore forensic hardware, software, storage devices, operating systems, and file systems.
5. To study metadata, password security, encryption, and hidden files.
6. To understand the technical challenges involved in digital evidence handling.
7. To identify commercial, open-source, and field forensic tools.
8. To understand the basic process of forensic imaging, data recovery, analysis, and reporting.

3. Step-by-Step Procedure

Step 1: Understand the Fundamentals of Digital Forensics

Digital forensics is a branch of forensic science that involves identifying, collecting, preserving, examining, analyzing, and presenting digital evidence in a manner that maintains its integrity and reliability.

Digital evidence may be obtained from computers, laptops, mobile phones, storage devices, networks, cloud platforms, and other electronic systems.

Main activities:

- Identification: Locate potential sources of digital evidence.
- Collection: Acquire relevant digital data using appropriate methods.
- Preservation: Protect evidence from modification or destruction.
- Examination: Extract and organize relevant information.
- Analysis: Interpret the evidence to determine what happened.
- Presentation: Prepare a report containing findings and supporting evidence.

Example: If a computer is suspected of being used for unauthorized access, an investigator may examine system logs, browser history, timestamps, downloaded files, and relevant network records to reconstruct the activity.

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Step 2: Study the Phases of Digital Forensics

2.1 Identification

Identify the incident, affected systems, possible evidence sources, and investigation scope. Record the device type, location, serial number, condition, and relevant identifiers.

2.2 Collection and Acquisition

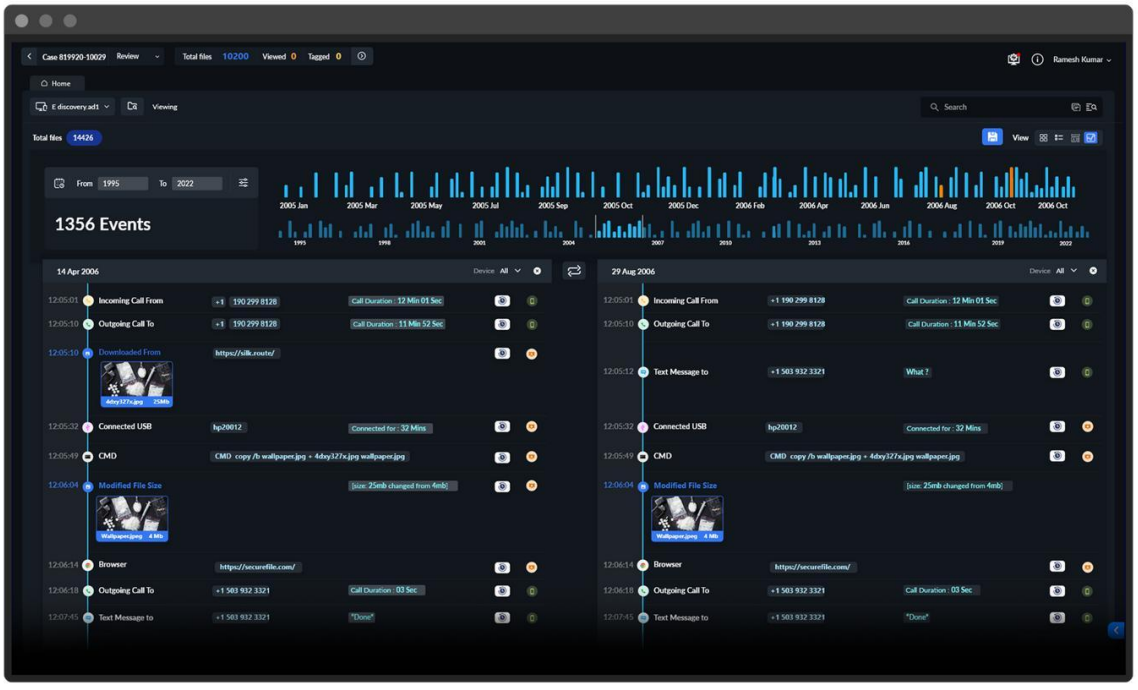
Collect relevant devices and acquire digital data using approved procedures. For storage media, create a forensic image where appropriate and document the acquisition method.

2.3 Preservation

Protect original evidence from alteration. Use a hardware or software write blocker when appropriate, calculate cryptographic hashes, maintain secure storage, and document every transfer of custody.

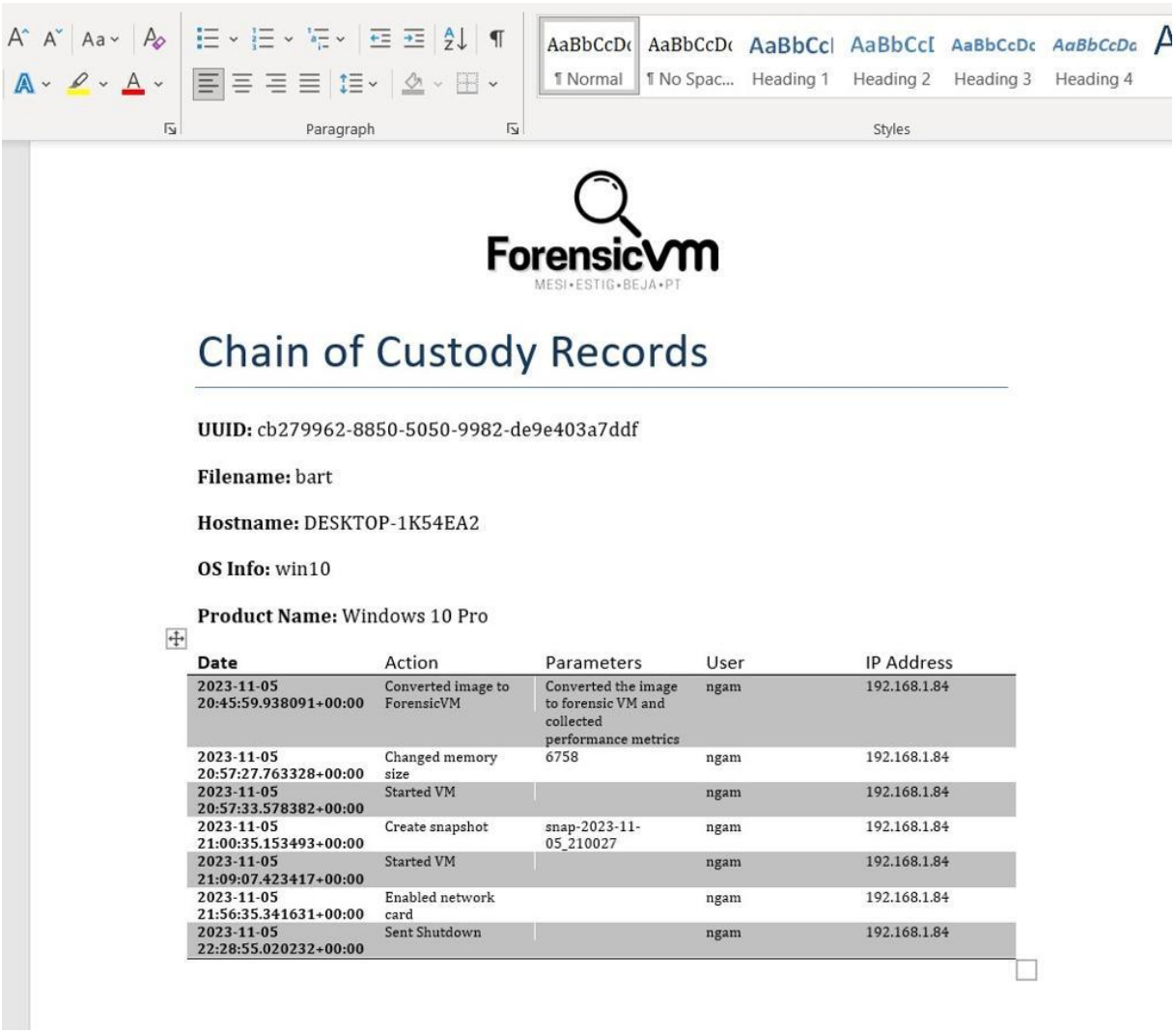
2.4 Examination

Use forensic tools to inspect files, partitions, metadata, deleted-file remnants, logs, and other relevant artifacts. Record the methods and tools used.



2.5 Analysis

Correlate timestamps, file artifacts, system logs, and other evidence to reconstruct events. Distinguish verified facts from assumptions and alternative explanations.



Paragraph Styles

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ForensicVM
MESI•ESTIG•BEJA•PT

Chain of Custody Records

UUID: cb279962-8850-5050-9982-de9e403a7ddf

Filename: bart

Hostname: DESKTOP-1K54EA2

OS Info: win10

Product Name: Windows 10 Pro

Date	Action	Parameters	User	IP Address
2023-11-05 20:45:59.938091+00:00	Converted image to ForensicVM	Converted the image to forensic VM and collected performance metrics	ngam	192.168.1.84
2023-11-05 20:57:27.763328+00:00	Changed memory size	6758	ngam	192.168.1.84
2023-11-05 20:57:33.578382+00:00	Started VM		ngam	192.168.1.84
2023-11-05 21:00:35.153493+00:00	Create snapshot	snap-2023-11-05_210027	ngam	192.168.1.84
2023-11-05 21:09:07.423417+00:00	Started VM		ngam	192.168.1.84
2023-11-05 21:56:35.341631+00:00	Enabled network card		ngam	192.168.1.84
2023-11-05 22:28:55.020232+00:00	Sent Shutdown		ngam	192.168.1.84

2.6 Documentation and Presentation

Prepare a clear report describing the evidence, acquisition details, tools, procedures, findings, limitations, and conclusions. Preserve the supporting records for independent review.

Step 3: Study the International Legal Framework of Cybercrime

Digital forensic investigations must comply with applicable laws, privacy requirements, authorization rules, and evidence-handling procedures.

Important legal frameworks include:

- Budapest Convention on Cybercrime: An international treaty addressing cybercrime, electronic evidence, and cooperation between countries. Its applicability depends on the country and its legal arrangements.
- United Nations initiatives on cybercrime: International work addressing cybercrime prevention, investigation, and cooperation.
- ISO/IEC 27037: Provides guidance for identifying, collecting, acquiring, and preserving digital evidence.
- ISO/IEC 27041: Addresses the suitability and adequacy of methods used in digital investigations.
- ISO/IEC 27042: Provides guidance on the analysis and interpretation of digital evidence.

- ISO/IEC 27043: Covers principles and processes for incident investigation.

Indian legal context:

- Information Technology Act, 2000, as amended.
- Bharatiya Sakshya Adhiniyam, 2023, including provisions relevant to electronic and digital records.
- Bharatiya Nagarik Suraksha Sanhita, 2023, for applicable criminal investigation and procedural requirements.
- Other applicable laws, rules, court directions, and organizational policies.

Procedure: Identify the jurisdiction, confirm legal authorization, document the scope of examination, protect privacy, maintain chain of custody, and follow applicable requirements for electronic evidence.

Step 4: Explore the Digital Forensics Environment

The forensic environment consists of hardware, software, storage media, and supporting facilities required to investigate digital evidence safely.

4.1 Forensic hardware

- Forensic workstation: A high-performance computer used to process evidence.
- Write blocker: Prevents write operations to a storage device during acquisition.
- Storage devices: External HDDs, SSDs, USB drives, and memory cards.
- Evidence storage: Secure media or repositories used to preserve forensic images.
- Network equipment: Switches, routers, and network capture devices when network evidence is relevant.

4.2 Forensic software

Forensic software helps investigators acquire images, recover files, examine file systems, search artifacts, analyze logs, and prepare reports.

Examples include Autopsy, FTK Imager, EnCase, FTK, and The Sleuth Kit.

4.3 Storage devices

Common storage devices include:

- Hard Disk Drive (HDD)
- Solid-State Drive (SSD)
- USB flash drive
- Secure Digital (SD) memory card
- Optical disc (CD/DVD)
- Mobile device storage
- Network-attached and cloud storage

Each storage type has different acquisition and recovery characteristics. For example, SSD TRIM operations may make deleted data unrecoverable, while cloud evidence may require provider cooperation and authorized account access.

4.4 Operating systems

- Windows: NTFS, FAT32, exFAT, Windows Registry, event logs, and Recycle Bin artifacts.
- Linux: ext4, XFS, system logs, shell history, and file permissions.
- macOS: APFS, HFS+, unified logs, and system metadata.
- Android and iOS: Application databases, device logs, messages, and other artifacts subject to device security and authorization.

4.5 File systems

A file system organizes files, directories, metadata, and storage allocation.

File system	Common use
FAT32	USB drives and older removable media
exFAT	Flash drives and large removable files
NTFS	Windows storage
ext4	Linux systems
APFS	Modern Apple devices
HFS+	Older macOS systems

Procedure: Identify the device, determine the file system, inspect the partition structure, and select a forensic tool compatible with the evidence format.

Step 5: Study Metadata, Password Security, Encryption, and Hidden Files

5.1 Metadata

Metadata is information describing a file or digital object.

Examples include:

- File name and extension
- File size
- Creation and modification timestamps
- File ownership and permissions
- EXIF information in photographs
- Document author and revision history
- Cryptographic hash values

Metadata helps establish context and timelines. However, timestamps can be altered, affected by time-zone differences, or misinterpreted, so they should be correlated with other evidence.

5.2 Password security

Passwords protect devices, accounts, and encrypted files. Investigators may examine password policy settings, authentication logs, and authorized password-recovery evidence.

Important concepts include password hashing, salting, multi-factor authentication, password complexity, and account lockout policies.

Password hashes are not the same as encrypted passwords: a secure password hash is designed for verification rather than direct decryption.

5.3 Encryption

Encryption converts readable information into ciphertext using a cryptographic key.

Common examples include AES-based file or disk encryption and full-disk encryption technologies such as BitLocker and FileVault.

Encryption can limit access to evidence. Investigators should use legally authorized credentials, recovery keys, or approved acquisition methods rather than attempting unauthorized access.

5.4 Hidden files

Hidden files are files that are not normally displayed in standard file browsing views.

Examples include hidden files in Windows and Linux, system files, and files stored in concealed directories or alternate data streams.

A hidden file is not automatically malicious. Investigators should inspect file attributes, paths, timestamps, hashes, and contents before drawing conclusions.

Procedure:

1. Open a forensic copy or disk image in an appropriate tool.
2. Inspect file properties and metadata.
3. Identify hidden files and directories.
4. Examine relevant file extensions and signatures.
5. Record hashes and timestamps.
6. Document significant findings without modifying the original evidence.

Result:

- Successfully understood the theoretical foundation and environment of digital forensics.
- Familiarized with multiple forensic tools commonly used by professionals.
- Gained an overview of evidence collection, the importance of forensically sound imaging, and challenges in analyzing digital evidence.
- Enhanced preparedness for practical forensic investigations by understanding the capabilities and applications of key forensic tools.